The Evidence Span Rarely Carries the Verdict: A Feasibility Study on Invasion-Biology Claims

[Author]

Abstract

We ask whether a black-box language model judges a short scientific claim by the sentence a human marked as evidence. From a CC BY full-text collection we keep 48 invasion-biology papers and write 67 claims, with the label rules fixed before any model is run. A retriever that searches only inside the source paper returns a chunk containing that evidence span for 40 of the 67 claims. Shown the claim and the five chunks, `gpt-4o-mini` matches the human label on 53 claims at temperature 0. Deleting the evidence span from those chunks changes 2 of the 40 labels, and both changes leave the human label. On the 32 claims that stay *supported*, sentence-level SHAP values are nonzero for at most ten sentences. Removing the highest-valued sentences moves the majority label off *supported*; removing the lowest-valued or randomly chosen sentences largely does not. The highest-valued sentence is only slightly closer, in the retriever's embedding space, to the evidence span than a random sentence from the same chunks. Five samples agree on 549 of 587 prompt settings. On the 67 original prompts, uncertainty is the mean token entropy of the full reply, averaged over the draws that share the majority label. It is 0.296 bits when that majority matches the human label and 0.463 bits when it does not. Writing the rationale before the label changes 6 of 67 labels and leaves the number of human-label matches at 53. A second call then sees the claim, the label, the rationale, and one or two quotes, but not the chunks. The label is kept only when the two calls agree and every quote is an exact substring of a retrieved chunk. One revision is allowed. This keeps 12 labels after one round, 11 of them matching the human label, and 19 after the revision, 15 of them matching. Every kept label is the temperature-0 label. The other claims are left unanswered, mostly because the quote is not an exact copy. The study is small, singly annotated, and limited to one model.

1 Introduction

A useful reading of a scientific paper is a judgment: a stated claim is supported by the paper, contradicted by it, or left open. For that judgment to be checkable, it has to depend on a passage that actually bears on the claim. Supplying a language model with retrieved passages does not by itself show that the passage a human would cite is the one that fixes the label. The model may repeat a nearby paraphrase, react to the topic of the chunks, or emit a label that a written rationale does not determine.

We run a small set of prompt checks on that judgment. Deleting the evidence span asks whether the label used that sentence. Sentence-level SHAP and further deletions ask which remaining sentences still carry a *supported* label. Repeated samples ask how stable the label is, and reversing the order of rationale and label asks whether the label is fixed before the rationale is written. We do not estimate a causal effect of a biological intervention. Deleting a span is an edit of the prompt.

Four questions organize the experiments. Does a same-paper retriever place the human evidence span in the top five chunks? After that span is removed, which remaining sentences still carry a *supported* label, and are those sentences restatements of the span? How stable is the label under five samples, and does writing the rationale first change it? Does a second call, which sees the quotes but not the chunks, keep the label, and does one revision raise the number of human-label matches?

The answers are narrow and mostly negative about the evidence span itself. The span is often retrieved, and the model often matches the human label, but deleting the span rarely changes the label. A few sentences with high SHAP values do change it. Those sentences are only marginally closer to the evidence span than other sentences from the same paper. Samples usually repeat one another. Mean token entropy is higher when the majority misses the human label, and the ranges overlap. Putting the rationale first makes a handful of labels more cautious and does not raise agreement with the human labels. A second call that does not see the chunks keeps 12 of the 67 labels after one round and 19 after one revision. Every kept label is the temperature-0 label. The number of human-label matches falls, because the other claims are left

unanswered. Most of those abstentions are quotes that are not exact substrings of the chunks.

2 Related work

Conditioning a generator on retrieved text is the idea behind retrieval-augmented generation (Lewis et al., 2020). Our setup is narrower. The index contains only the paper that the claim was written from, and the query is filtered to that paper's DOI. The experiment therefore measures whether the evidence span is among the five chunks most similar to the claim inside its own paper. It does not measure open-corpus retrieval.

Faithfulness, in the sense of Jacovi and Goldberg (2020), is a property of an explanation relative to the model's own prediction, not relative to a human label. We use two operational checks. The first deletes input sentences and asks whether the label moves. The second changes the order in which the rationale and the label are generated, following generation-order tests in vision-language models (Liang et al., 2026). A label that stays the same when the cited span is removed is not, on that check, fixed by the span. A label that changes when the rationale must be written first is not fixed before the rationale is produced.

SHAP assigns an additive importance to each input feature of one prediction (Lundberg and Lee, 2017). We use KernelSHAP with sentences as features and with the log probability of the label token *supported* as the value function. The resulting values are an explanation of that score under a sparsity penalty. They are not a unique account of every token the model might have used.

Sampling several answers and keeping the majority is the idea behind self-consistency (Wang et al., 2023). Agreement, the fraction of samples that repeat the majority, is used as a stability measure by Kuhn et al. (2023). Our uncertainty is the Shannon entropy of each completion token, averaged over the tokens of a reply and then over the draws that emitted the majority label. We do not build a risk-coverage curve.

In invasion biology, D'Souza et al. (2025) use GPT-4o without domain fine-tuning to extract species, locations, habitats, and ecosystems from a large literature set. The full texts we label are taken from the companion text-mining release of that line of work (D'Souza, 2024). Their task is entity extraction. Ours is a three-way judgment of a claim against a small set of chunks from the same paper.

3 Data

3.1 Papers

The source file is `publications_tdm.csv` from D'Souza (2024). It has no license column of its own. Candidate papers are those with a non-empty full text whose version of record OpenAlex records as CC BY (Priem et al., 2022). A paper enters the pool only if OpenAlex types it as an English article in the published version, the full text has at least 20,000 characters, and the title signals invasion biology or the journal is one of five specialist venues.[1] Reviews, meta-analyses, editorials, opinions, syntheses, frameworks, corrections, and a short list of off-topic title patterns are excluded, as are full texts whose opening is a review header or a horizon scan.

Themes are assigned from the first matching pattern in the title: freshwater, else plant, else marine, else insect, else other. Only journals with at least five qualifying papers are kept. Within each theme, papers are taken from the most recent years, alternating across journals, with at most three papers per journal. A paper stays in the set only if CC BY, without NC, ND, or SA, is readable on the publisher's page or in the publisher PDF. Every one of the 48 retained papers was checked this way. One paper is CC BY 3.0 rather than 4.0 (`10.1002/ecs2.2097`). The set has 48 papers, 31 journals, and at most two papers per journal: 12 freshwater, 12 plant, 8 marine, 8 insect, and 8 other.

The CSV text is an extract of publisher PDF text, not a clean transcription. A spot check of 11 papers, across ten publishers, found a mid-document passage from the CSV inside the version of record or the publisher PDF. None of the 48 extracts carries an accepted-manuscript or preprint header, and each contains its own DOI. Words are often glued together, running headers sit in the body, and two papers carry a library download stamp. Evidence spans are therefore exact substrings of the CSV, including its defects, so that a later deletion can find them. They are not cleaned quotations.

3.2 Claims

The annotation rules were fixed before the first label. Each row is one claim about a finding in one

[1] Aquatic Invasions, BioInvasions Records, Biological Invasions, NeoBiota, and Management of Biological Invasions.

of the 48 papers: spread, establishment, impact, detection, or management. Methods, definitions, and another paper's result quoted as if it were this paper's finding are out of scope. At most two claims are taken from a paper, and the second only when it concerns a different finding. Every paper contributes at least one claim. The claim and the label are in English.

The evidence span is a contiguous piece of the CSV full text. The label is attached to that span. *Supported* means the span reports the paper's own finding and the finding matches the claim. *Not supported* means the paper tests the claim and the span reports the opposite or no effect. *Unclear* means the span bears on the claim but does not settle it: the paper reports conflicting results, a thin sample, or says that the finding is open. Speculation that appears only in the introduction, with no finding later in the paper, is not labeled. Negative and open findings are included on purpose.

One annotator applied the rules. The file contains 67 claims: 46 *supported*, 13 *not supported*, and 8 *unclear*. Each evidence span occurs verbatim in the full-text field.

4 Methods

4.1 Retriever

Each of the 48 full texts is split with a recursive character splitter into chunks of 1,000 characters with an overlap of 200. Chunks are embedded with `text-embedding-ada-002`, the default embedding model of the library version used here, and stored with the paper DOI as metadata. The query is the claim. Search is filtered to that DOI and returns five chunks. The gold label and the evidence span are not part of the query. A hit is an exact substring match of the evidence span inside one of the five chunks. A chunk that does not contain the span can still rank above one that does, because the embedding compares the whole chunk with the claim.

4.2 Label

The labeler is `gpt-4o-mini` at temperature 0. The user message contains the claim and the five chunks. The system message asks for JSON only, with the label before the rationale. The label is one of *supported*, *not supported*, and *unclear*, and the rationale is one or two sentences. The instructions define the three labels in the same terms as the annotation rules and tell the model to use only the supplied passages. The gold label and the evidence span are not shown.

4.3 Evidence deletion

The deletion is run only for the 40 claims whose evidence span sits in the top five chunks. Wherever the span occurs in those chunks, it is replaced by a space. The same labeler then sees the claim and the edited chunks. The comparison is with the previous prediction, not with the gold label. A span that the retriever never returned is not deleted, because the model did not see it in the first place.

4.4 Sentence SHAP

Sentence SHAP is restricted to the claims whose predicted label is still *supported* after the evidence span has been removed. There are 32 such claims. Sentences are the units. They are split on sentence-ending punctuation, with abbreviations, decimals, and fragments shorter than 40 characters folded into a neighbor. The value function is the log probability that the generated label starts with the token *supported*. That probability is read from the token log-probabilities at temperature 0, with the five most likely tokens returned. If *supported* is not among them, the score is placed below the lowest returned alternative. The background passed to KernelSHAP is an empty mask, rendered to the model as no text. The explained instance keeps every sentence. The number of sampled coalitions is twice the number of sentences, and at least 40. An ℓ_1 penalty, `num_features(10)`, keeps at most ten sentences away from zero. The random seed for each claim is fixed from the claim identifier.

Three deletion sets are then built from these values, at sizes 1, 3, 5, 7, and 10. The highest set keeps only sentences with a positive value, in descending order, and stops early if fewer than the requested number are positive. The lowest set sorts by ascending value. The random set sorts by a draw that is fixed per claim. The same draw is used at every size, so the smaller sets sit inside the larger ones. The same is true of the lowest set. Deleted sentences are removed from the chunks that remain after the evidence span has already been removed, and the labeler is called again at temperature 0.

4.5 Span similarity

Each selected sentence and each evidence span is embedded with the same model as the retriever. The score is cosine similarity. For each claim and

each set, we average the similarities of the sentences in the set, then average those claim-level means across the 32 claims. The comparison that matters is inside a claim: the highest set against the lowest set and against the random set, which are drawn from the same chunks and therefore share the paper's topic.

4.6 Sampling

Every prompt that had been labeled once is labeled five more times, at temperature 0.7 and top-p 0.9. Each draw is capped at 24 tokens, which is long enough for the label field. The reported label is the majority label. If two labels tie for the highest count, the one that appears first in the five draws is kept. That tie occurs once. Agreement is the size of the majority divided by five. Uncertainty is computed only for the 67 original prompts. Five further draws, at the same temperature and top-p, generate the whole reply. It is the mean Shannon entropy, in bits, of every token in a reply, averaged over the draws that share the majority label. Each token uses its twenty most likely alternatives, and the remaining probability is one extra outcome. The sentence sets are not recomputed. SHAP is not rerun. The 32-claim subset stays the subset defined by the temperature-0 deletion.

4.7 Rationale first

On the original 67 prompts, at temperature 0 and with no token cap, the JSON schema is reversed: the model must write the rationale first and the label second. The passages are unchanged. A response counts as ordered correctly when the rationale key precedes the label key in the raw JSON. The comparison is with the temperature-0 label from the original schema.

4.8 Second reader

The temperature-0 labeler is called again on the same 67 claims and the same five chunks, still at temperature 0. The reply adds one or two quotes to the label and the rationale. A quote is supposed to be a contiguous piece of a retrieved chunk. A second call sees the claim, the proposed label, the rationale, and the quotes. It does not see the chunks. It returns its own label, under the same three definitions.

The label is kept when the two labels agree and every quote is an exact substring of one of the five chunks. Otherwise the first call is repeated once. That repetition still sees the chunks, and it is shown the second label, the second rationale, and the quotes that were not substrings. The second call then runs again. If the labels still disagree, or a quote is still not a substring, the claim is left unanswered. The kept label is the label from the call that saw the chunks. A one-round setting skips the repetition and leaves the claim unanswered at the first miss. The human label is shown to neither call.

Table 1: Evidence span found inside the top five chunks, by human label.

Human label	Hit	Miss	Claims
Supported	32	14	46
Not supported	3	10	13
Unclear	5	3	8
All	40	27	67

Table 2: Temperature-0 labels against the human label. Rows are the human label.

Human label	Not supported	Supported	Unclear
Not supported	10	3	0
Supported	4	42	0
Unclear	6	1	1

5 Results

5.1 Retrieval and the label

The evidence span is in the top five chunks for 40 of 67 claims. Table 1 splits the hits by the human label. *Not supported* is rarely retrieved. The claim states a finding that the evidence span denies, so a chunk that mentions the finding can outrank the span that denies it.

At temperature 0 the predicted label matches the human label on 53 of 67 claims (Table 2). Three *not supported* claims are predicted *supported*. Four *supported* claims are predicted *not supported*. Six of the eight *unclear* claims are predicted *not supported*.

5.2 Evidence deletion

Of the 40 claims with a retrieved span, 38 keep the temperature-0 label after the span is deleted. Two change, and both were matches to the human label before the deletion. Claim c010 moves from *not supported* to *supported*. Claim c062 moves from *supported* to *not supported*. In those two prompts the span was necessary for the label the model had produced. In the other 38, other text in the five chunks is enough to keep the label.

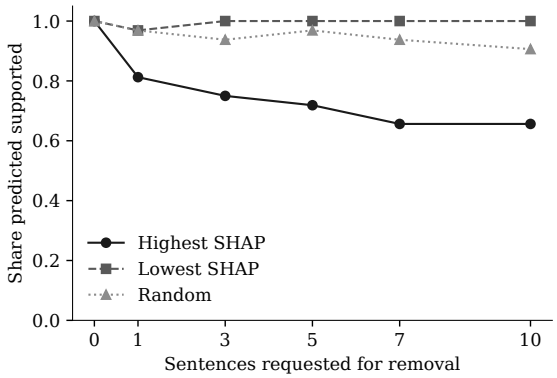


Figure 1: Share of the 32 claims still labeled *supported* by the majority of five samples, after removing sentences from the chunks that no longer contain the evidence span. At ten sentences the highest set contains 9.44 sentences on average.

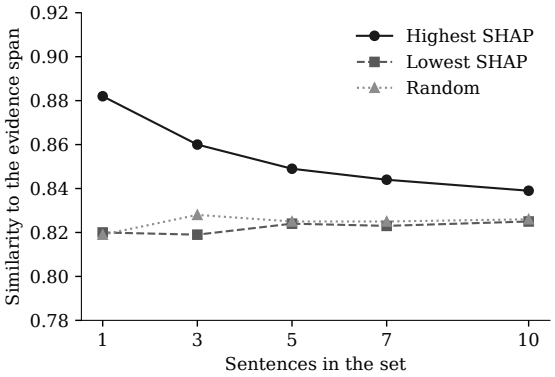


Figure 2: Mean cosine similarity between the evidence span and the sentences chosen for deletion. The vertical axis starts at 0.78. All three curves lie above 0.81 because the sentences come from the same paper as the span.

5.3 What carries the label

The 32 claims that stay *supported* contain 1,009 sentences after the span is removed, between 20 and 41 sentences per claim. The sparsity penalty leaves 7 to 10 sentences positive and 0 to 3 sentences negative. The remaining sentences, 10 to 31 per claim, are zero. One of the 32 claims, c035, has the human label *unclear*, so a *supported* prediction matches the human label on 31 of the 32.

Figure 1 plots the share of these 32 claims whose majority label is still *supported* after sentences are removed. Before any SHAP sentence is removed, all 32 majority labels are *supported*. Removing the single highest sentence leaves 26. Removing ten highest sentences, 9.44 on average because only positive sentences are eligible, leaves 21. Removing lowest or random sentences leaves 29 to 32. The lowest curve is not monotone: one claim flips at size 1 and is *supported* again at size 3. The same direction appears under temperature-0 decoding. On 471 of the 480 removal prompts, the majority label equals the temperature-0 label. Agreement falls as high-valued sentences are removed and stays near 1 when the removed sentences have low values. The full per-condition counts, the sentence values, and the sample draws are in the project repository.[2]

5.4 Restating the span

They are only slightly closer to it than other sentences from the same chunks. Every sentence in those chunks is already near the evidence span: cosine similarity averages 0.826, with a minimum of 0.689 and a maximum of 0.965. Figure 2 isolates the small gap. The single highest sentence averages 0.882, against 0.819 for one random sentence and 0.820 for the lowest sentence. The highest sentence is closer than the random sentence in 27 of 32 claims. The mean gap shrinks to 0.014 once ten sentences are included. Within a claim, the rank correlation between the SHAP value and the similarity averages 0.193, and it is positive for 27 of 32 claims.

The average hides two different cases. For c001 the highest sentence repeats the evidence span with one lexical change, algal biomass in place of productivity, at similarity 0.965 against 0.821 for the random sentence. Figure 3 shows every nonzero value for that claim. For c005 the highest sentence is a truncated fragment about the limit of detection, at similarity 0.826, while the lowest sentence, which states where *Aphanomyces astaci* was detected, sits at 0.929. A high SHAP value can mark a restatement of the finding, and it can mark a broken piece of the extract that is not the finding.

5.5 Stability of the label

Five draws were taken for 587 prompts, 2,935 draws in total. On the 67 original prompts the majority label matches the temperature-0 label 66 times. The exception is c056, human label *supported*, temperature-0 label *supported*, majority *unclear*, agreement 0.6. The majority therefore matches the human label on 52 of 67 claims, one fewer than the single decode. The three *not supported* claims predicted *supported* are the same three as at temperature 0. After the evidence span

[2] https://github.com/Acul01/sable

Table 3: The six claims whose label changes when the rationale is written before the label. Both decodes use temperature 0 and the same five chunks.

Claim	Human label	Label first	Rationale first
c046	Unclear	Unclear	Not supported
c056	Supported	Supported	Unclear
c060	Unclear	Not supported	Unclear
c064	Not supported	Supported	Unclear
c066	Supported	Not supported	Unclear
c067	Unclear	Not supported	Unclear

is removed, the majority matches the temperature-0 ablated label on all 40 claims, and c010 and c062 are again the claims that flip.

Agreement equals 1 on 549 of the 587 prompts. Only 3 of the 67 original prompts have agreement below 1. On those 67 prompts, the uncertainty of the full reply averages 0.296 bits when the majority matches the human label and 0.463 bits when it does not. All 14 mismatches are unanimous, and the two ranges overlap.

5.6 Rationale first

All 67 responses place the rationale key before the label key. Six labels change relative to the original schema (Table 3). The number of matches to the human label stays 53. Two claims become correct, c060 and c067. Two become incorrect, c046 and c056. Two change and remain incorrect, c064 and c066. Five of the six new labels are *unclear*. Requiring the rationale first makes the label slightly more cautious. It does not sort correct from incorrect labels.

5.7 Second reader

Of the 67 claims, the one-round setting keeps 12 labels and leaves 55 unanswered (Table 4). Eleven of the 12 match the human label. The one-revision setting keeps 19, of which 15 match the human label. Seven of those 19 were left unanswered after one round, and four of the seven match the human label. No label kept after one round is withdrawn. Every kept label, in either setting, is the temperature-0 label. Neither setting corrects a temperature-0 label that missed the human label. After one round, 42 of the unanswered claims are ones the temperature-0 label got right, and 13 are ones it got wrong. After the revision those counts are 38 and 10.

The unanswered claims are mostly quotes. After one round, 48 claims have a quote that is not an exact substring of a retrieved chunk. On 38 of the 55 unanswered claims the two labels agree, and the quote is what drops the claim. The two labels disagree on 17 claims, and on 7 of those the quote is an exact substring. The revision leaves the same pattern: 46 claims still have a quote that is not a substring, and the labels disagree on an exact quote for 2 claims.

Table 4: Labels kept when a second call sees the quotes but not the chunks. Match counts agreement with the human label. The temperature-0 row answers every claim.

Setting	Answers	Match	Unanswered
Temperature 0	67	53	0
One round	12	11	55
One revision	19	15	48

6 Discussion

The evidence span is a weak lever on this labeler. It is missing from the top five chunks in 27 of 67 claims, especially when the human label is *not supported*, and deleting it moves only 2 of the 40 labels that were in a position to move. The sentences that do move a *supported* label are the ones with the highest SHAP values. Lowest-valued and random deletions of the same size do not. That contrast is the useful content of the attribution: the label is concentrated, and the concentration is not an artifact of deleting any sentence.

Concentration is not the same thing as restating the evidence span. Inside one paper, almost every retrieved sentence is close to the span in embedding space, so a cosine of 0.82 does not mean that a sentence carries the finding. The extra similarity of the top sentence is real and small, and the c001 and c005 cases show both a paraphrase and a fragment that the extractor damaged. A SHAP ranking on this text should be read against the sentence, not against the number alone.

Sampling adds a stability check more than a correctness check. The majority reproduces the single decode, and the draws agree even on labels that miss the human annotation. Mean token entropy is higher on those misses, 0.463 bits against 0.296, and the ranges overlap. Putting the rationale first is a faithfulness check of a different kind. The label token is then conditioned on text the model has just produced. The effect is a shift toward *unclear* in a few claims, with gains and losses that cancel in the match count.

The second reader does not raise the number of human-label matches. It keeps a label by re-

peating the temperature-0 label, and only when a quote is copied exactly and the two calls agree. That filter leaves unanswered most of the claims the temperature-0 label got right, and most of the claims it missed. The quotes that fail are often close to a sentence in the chunk: a shortened clause, a paraphrase, or an ellipsis. The check counts only matching characters. Where the characters do match, disagreement is uncommon: seven claims after one round, and two after the revision.

None of these edits is a causal estimate about an invasion. They are edits of a prompt. They also use a retriever that is told which paper to search. A system that had to find the paper would face a harder problem than the one measured here.

7 Conclusion

On 67 invasion-biology claims, a same-paper retriever and `gpt-4o-mini` often recover the human label, and they often do so without the sentence the annotator marked. The label that remains is carried by a small set of high-SHAP sentences. Those sentences are only a little closer to the evidence span than the other sentences in the retrieved chunks. Five samples usually agree, including when the agreed label is wrong. Mean token entropy is higher on the misses, and the ranges overlap. Writing the rationale first changes six labels and not the match count. A second reader who sees the quotes but not the chunks keeps that label on 12 claims, or on 19 if one revision is allowed, and leaves the rest unanswered. The kept labels are the temperature-0 labels, and the match count falls. The study says where this particular label comes from in the prompt. It does not say that the model has checked the paper.

Limitations

There are 67 claims, written by one annotator, each tied to a single span. A second span in the same paper could support a different label. The spans keep the defects of the PDF extract, and those defects enter both the deletion and the SHAP sentences. The search never leaves the source paper. SHAP explains one score, the log probability of the token *supported*, under a penalty that zeros most sentences, so the lowest ranks are mostly zeros plus a few negative values. Uncertainty is the mean token entropy of the full reply on the 67 original prompts. Each token uses twenty alternatives and one residual bin, so the value is a lower bound on the entropy of the full vocabulary. The five draws are one sample at one temperature, not a variance across seeds. One model, `gpt-4o-mini`, was used. The second reader sees the quotes but not the remaining chunks, and a quote counts only as an exact substring, so a paraphrase is an abstention. The revision is one extra round. Risk-coverage curves and an open-weight model were left out of the study on purpose.

References

Jennifer D'Souza. 2024. Invasion biology WikiProject scientific papers: Text data mining and LLM-based information extraction of species, locations, habitats, and ecosystems.

Jennifer D'Souza, Zachary Laubach, Tarek Al Mustafa, Sina Zarrieß, Robert Frühstückl, and Phyllis Illari. 2025. Mining for species, locations, habitats, and ecosystems from scientific papers in invasion biology: A large-scale exploratory study with large language models. In *Proceedings of the 1st Workshop on Ecology, Environment, and Natural Language Processing (NLP4Ecology2025)*, pages 16–23, Tallinn, Estonia. University of Tartu Library.

Alon Jacovi and Yoav Goldberg. 2020. Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4198–4205.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, et al. 2022. Language models (mostly) know what they know. *Preprint*, arXiv:2207.05221.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations*.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33.

Siting Liang, Luca Rippe, Omar Adjali, and Daniel Sontag. 2026. Evaluating explanation-driven vision–language reasoning via generation order interventions. In *Proceedings of the IJCAI-ECAI 2026 Workshop on Explainable Artificial Intelligence*, pages 203–209, Bremen, Germany.

Scott M. Lundberg and Su-In Lee. 2017. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30.

Jason Priem, Heather Piwowar, and Richard Orr. 2022. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. In *Proceedings of the 26th International Conference on Science and Technology Indicators (STI 2022)*.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*.

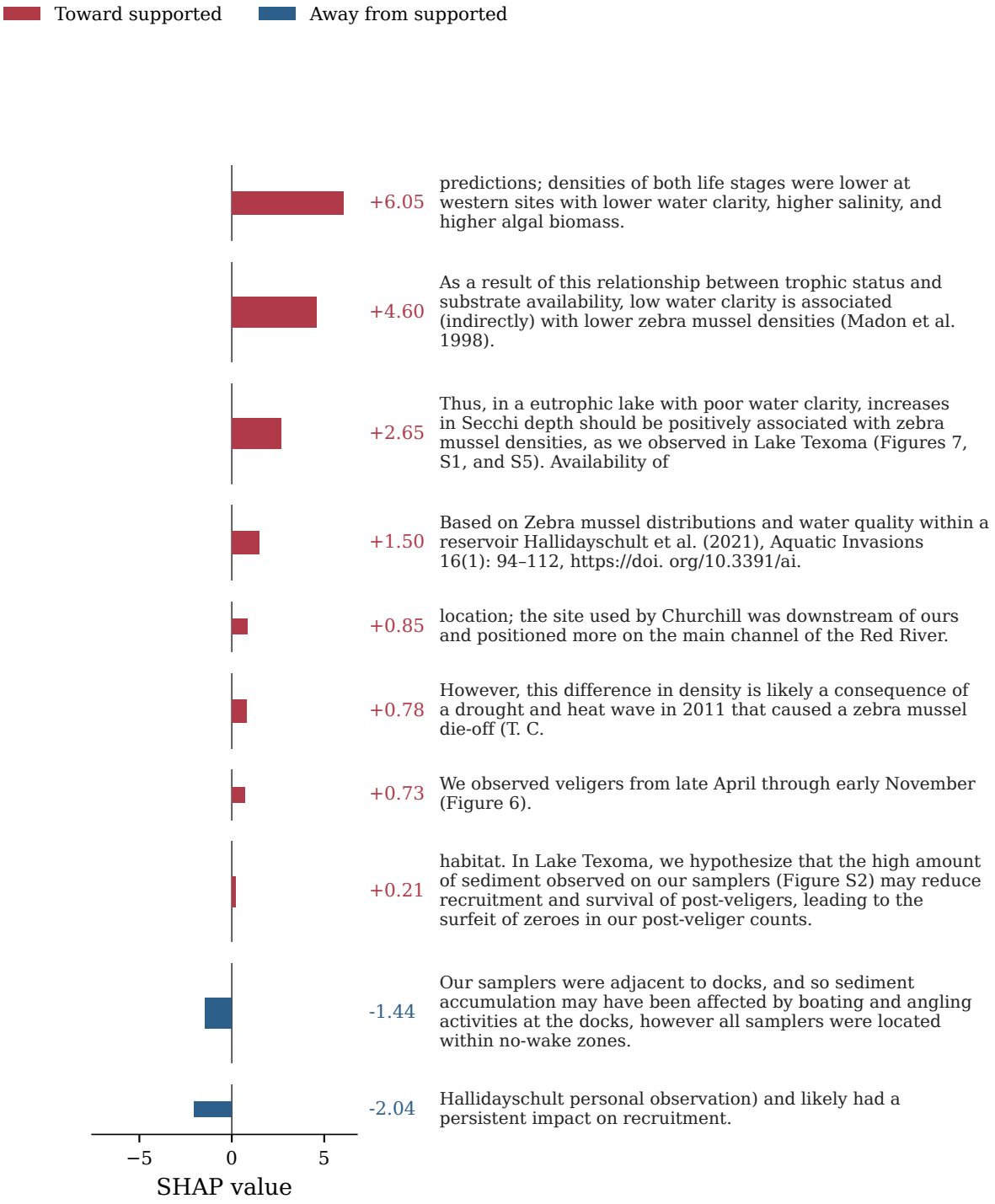


Figure 3: Nonzero sentence SHAP values for claim c001 after the evidence span was removed. Red raises the log probability of *supported*; blue lowers it. Seventeen of the 27 sentences are zero and are omitted. The sentence at $+6.05$ is the near-paraphrase of the deleted span.